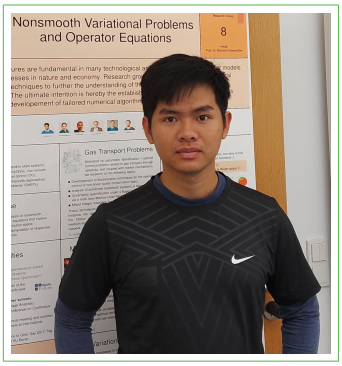
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ResearchGate/Hong Quan Ba Nguyen



Hong Nguyen

Curriculum Vitae

"The essence of mathematics lies in its freedom." - Georg Cantor
"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Profile

Applied Mathematician specializing in continuous physical systems (PDEs, fluid dynamics) and mathematical optimization. Highly motivated to bridge rigorous numerical analysis with data-driven operator learning to model and control complex nonlinear dynamical systems. Proven research experience in reduced-order modeling (ROMSOC) and independent theoretical research.

Education

2020–2021 **PhD student enrolled in Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis & Stochastics* (WIAS), Berlin, Germany.
Within the European Innovative Training Network **ROMSOC** (*Reduced Order Modelling, Simulation & Optimization of Coupled Systems*). Focused on shape optimization, fluid mechanics (Navier-Stokes), and reduced-order modeling for continuous physical systems.

2018–2019 **Master de Sciences, Technologie, Santé, Mathematiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).

2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis & Numerical Analysis.

2011–2014 **Elementary & Olympiad Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

Title *Staggered & well-balanced discretization of shallow-water equations*. [pdf][slide]

Supervisor Prof. Nicolas Seguin

Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Master Courses

2018–2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications, 2018–2019:**

Sep 2018 Prof. Nicoletta Tchou. *Sobolev spaces & elliptic equation*.

Sep 2018 Prof. Florian Méhats. *Hyperbolic equations*.

Nov 2018 Prof. Nicolas Seguin & Eric Darrigrand. *Finite Element Method* (FEM).

Nov 2018 Prof. Nicolas Seguin. *Numerics of transport*.

Nov 2018 Prof. Mohammed Lemou. *Numerical methods for kinetic transport equations*.

Dec 2018 Prof. Luis Miguel Rodrigues. *Master 2 Seminar*
▷ Hong Nguyen. "On the smallest constant for a Gagliardo-Nirenberg functional inequality". [pdf][slide]

Mar 2019 Vincent Duchêne. *Modeling & mathematical analysis of shallow free-surface fluid flows*.

Mar 2019 Prof. Marc Quincampoix. *Optimal control & Hamilton-Jacobi equations*.

Research Interests

- **Nonlinear Dynamical Systems** ▷ Spectral Submanifolds (SSM), Fluid-Structure Interactions, Model Reduction.
- **Data-Driven Modeling & Control** ▷ Operator Learning, Scientific Machine Learning (SciML), Physics-Informed Neural Networks (PINNs).
- **Mathematical Analysis & Fluid Mechanics** ▷ Navier–Stokes equations, Computational Fluid Dynamics (CFDs), Turbulence modeling.
- **Mathematical Optimization** ▷ Optimal Control, Shape & Topology Optimization.
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).

Research & Professional Experience

Vocational & Research

Jan 2025–now **Lecturer**, *University of Management & Technology (UMT HCMC)*, Ho Chi Minh City.
School of Technology (SOT): expertise: Mathematics for Data Science & Artificial Intelligence.

Jan 2025–now **Mathematical Olympiad trainer**, *UMT HCMC*, Ho Chi Minh City.
Training undergraduate students for Vietnamese Mathematical Competitions for Undergraduate Students.

Jan 2025–now **Olympiad in Informatics trainer**, *UMT HCMC*, Ho Chi Minh City.
Training undergraduate students for Vietnamese Olympiad in Informatics for Undergraduate Students.

Jan 2022–Dec 2024 **Independent Researcher**, Ben Tre city, Vietnam.
Conducted independent research focusing on applied mathematics, optimization, and algorithmic reduction. Authored and submitted multiple preprints bridging theoretical mathematics with computational applications.

Apr 2021–Dec 2021 **Software Development Engineer (SDE)**, *Math$_+$ Tec GmbH*, Vienna, Austria.
Secondment for technical & scientific training as part of the ROMSOC project. Applied model reduction techniques in an industrial context.

2020–2021 **Marie Curie PhD Researcher**, *Weierstrass Institute (WIAS)*, Berlin, Germany.
Marie Skłodowska-Curie fellow within the **ROMSOC** network. Investigated optimal shape design and model reduction of coupled systems.

Mar–Jun 2019 **Master Internship**, *Institut de recherche mathématique de Rennes – IRMAR*, Rennes, France.

Miscellaneous

2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students* (CBT), Vietnam.
Trainer of the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Peer-Reviewed Journal Publications

[1] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. *"Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term"*. *Advances in Nonlinear Studies* 21(1), pp. 1–22 (2021). DOI: 10.1515/ans-2020-2076.

Manuscripts Under Review / Submitted

[s1] Nguyen Quan Ba Hong. *"Universal Gradient Estimates and Exact Free Boundary Profiles for the Singular Porous Medium Equation"*. [Submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editorial assignment pending] (extending NQBH's bachelor thesis).

[s2] Nguyen Quan Ba Hong. *"Spatial Decoupling and Complete Quenching for Singular Porous Medium Equations"*. [Submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editorial assignment pending] (extending NQBH's bachelor thesis).

[s3] Nguyen Quan Ba Hong. *"An Automated Resolution and Generalization of Rowland's Binomial Divisibility Conjectures via Algebraic Rank Minimization"*. [resubmitting].

[s4] Nguyen Quan Ba Hong. *"Alexandrov Completions, Simplicial Nerves, and Ahlswede–Zhang Obstructions over Finite Covering Spaces"*. [resubmitting], [codes].

[s5] Nguyen Quan Ba Hong. *"Uniform Algebraic Stratification and Automated Refutation of Elementary Symmetric Inequalities via Invariant Positivstellensatz Certificates"*. [Submitted to Journal of Symbolic Computation (JSC), status: under review].

[s6] Nguyen Quan Ba Hong. *"Boundary Optimization for General Overlapping Coverings via Extremal Combinatorics"*. [resubmitting].

Selected Working Papers

[w1] Nguyen Quan Ba Hong. *"Convergence and Asymptotic Commutativity of the Adjoint PISO Algorithm on Polyhedral Meshes"*. `[In preparation]`.

[w2] Nguyen Quan Ba Hong. *"The Functional Level-Set Slack Identity: Elevating Empirical OOD Baselines into Exact Continuous Combinatorial Equations"*. `[In preparation]`.

[w3] Nguyen Quan Ba Hong. *"Global Convergence and Lax–Wendroff Consistency of the Offset Equilibrium Staggered Scheme for Shallow Water Equations at Dry-Wet Transitions"*. `[In preparation]` (extending NQBH's master thesis).

Books & Other Writings

[book1] Nguyen Quan Ba Hong. *"Theoretical Competitive Programming (Olympiad in Informatics & ACM-ICPC)"*.

[book2] Nguyen Quan Ba Hong. *"Programmable Mathematical Olympiads (VMC & High School)"*.

Courses

Oct 12–23, 2020 Thematic Einstein Semester (TES): Energy-based mathematical methods for reactive multiphase flows.
▷ TES Student Compact Course: *Variational Methods for Fluids & Solids*.

Apr–Jun 2020 Dr. Stephan Schmidt. *Computational Geometry & Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

Seminars

2020–2021 Berliner Oberseminar Nichtlineare partielle Differentialgleichungen (Langenbach-Seminar), Research Group 1: *Partial Differential Equations*, WIAS Berlin, Germany.

Mar 2020–now Material Modeling Seminar, WIAS Berlin, Germany.

2020–2021 Numerical Mathematics Seminars, Research Group 3: *Numerical Mathematics & Scientific Computing*, WIAS Berlin, Germany.

2020–2021 Modern Methods in Applied Stochastics & Nonparametric Statistics, Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*, WIAS Berlin, Germany.

2020–2021 Joint Research Seminar on Mathematical Optimization, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

2021–2021 Optimization, Control & Inverse Problems, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

2020–2021 Berlin Mathematics Research Center MATH$^+$ Events.

2020–Apr 2021 ROMSOC Scientific ESRs Seminar.

Schools

Jul 26–30, 2021 Summer Schools on Applied Mathematics and Mechanics 2021 (SAMM 2021): *Shape & Topology Optimization*. GAMM Juniors, Graz University of Technology (Technische Universität Graz), Graz, Austria.

Jul 1–2, 2021 Durham Days of Analysis & PDE Summer School. Durham University, United Kingdom.

Sep 7–10, 2020 Model Order Reduction Summer School 2020 (MORSS2020). École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Jul 20–24, 2020 162nd European Study Group with Industry. University of Leeds, United Kingdom.

Workshops

Apr 29, 2021 FUHRI2021: Finite volUme metHods for Real-world applIcations, WIAS Berlin, Germany.

Sep 28–29, 2020 ROMSOC Internal Workshop.

Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

Conferences

Mar 24–25, 2021 Annual Meeting of the Second Phase of the DFG Priority Programme 1962 (SPP1962): *Non-smooth & Complementarity-based Distributed Parameter Systems: Simulation & Hierarchical Optimization*. Deutsche Forschungsgemeinschaft (DFG) + WIAS, Germany.

Sep 14–17, 2020 Deutsche Mathematiker-Vereinigung (German Mathematical Society) Annual Meeting 2020 (DMV2020), Technische Universität Chemnitz, Chemnitz, Germany.

Talks

Apr 12, 2021 Hồng Quản Bá Nguyễn (WIAS Berlin). "*Optimal Shape Design of Air Ducts in Combustion Engines: Design a General Shape Optimization Framework*". ROMSOC ESR Seminar. [abstract]

Sep 7, 2020 Hong Nguyen (WIAS Berlin). "*A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes*". Model Order Reduction Summer School 2020 (MORSS2020), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. [abstract][slide]

Jun 26, 2019 Hong Nguyen. "*Staggered & well-balanced discretization of shallow water equations*". PhD interview, Department of Mathematics, RWTH Aachen University, Aachen, Germany. [slide]

Miscellaneous

Feb 22, 2021 WIAS-Day2021 ▷ Hong Quan Ba Nguyen (Research Group 8)'s Research Profile.

Nov 30, 2020 ROMSOC Blog ▷ Hong Nguyen. *Optimal shape design of air ducts* [pdf].

Jun 18, 2020 Prof. Luis Miguel Rodrigues's recognition of Master equivalence & admission to doctoral studies of NGUYEN Quan Ba Hong. IRMAR, Rennes, France.

May 27, 2019 Prof. Luis Miguel Rodrigues's recommendation letter for NGUYEN Quan Ba Hong. IRMAR, Rennes, France.

Visits

Apr 2021–Nov, 2021 MATH$_+$TEC GmbH, Vienna, Austria.

Apr 2021–Nov 2021 Vienna University of Technology/Technische Universität Wien, Vienna, Austria.

Apr 23, 2021 University of Vienna/Universität Wien, Vienna, Austria.

Mar 2020–Nov 2021 Humboldt-Universität zu Berlin, Berlin, Germany.

Jun 25–27, 2019 Department of Mathematics, Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen University, Aachen, Germany.

Sep 2018–Jun 2019 UFR Mathématiques, Université de Rennes 1, Rennes, Bretagne, France.

Sep 2018–Dec 2018 Université Rennes 2 - UFR Sciences sociales, Rennes, Bretagne, France.

Mar 2017 Vietnam Institute for Advanced Studies in Mathematics (VIASM), Hanoi, Vietnam.

Awards & Scholarships

Mar 2020–Aug 2021 Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship, *Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs*, Grant Agreement number: 765374 - Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC) [H2020-MSCA-ITN-2017-EID/ROMSOC/ESR11].

2018–2019 Lebesgue Master Scholarship for *Master 2* program within ENS Rennes program.

2017 Scholarship of *Vietnam Institute for Advanced Studies in Mathematics* (VIASM).

2016 Special Prize ⇐ First Prize in Algebra + First Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2016).

2015 Second Prize in Algebra, Second Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2015).

2014 First Prize #1 in Ben Tre city Mathematical Olympiad for High School Students.

2013 Second Prize in Ben Tre city Mathematical Olympiad for High School Students.

2012 Gold Medal ranked #2, in *Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam*.

Computer Skills

GitHub GitHub/NQBH ▷ GitHub/ROMSOC project.

Basic R, Fortran, Fireshape, Firedrake, FEniCS, STAR-CCM+.

Intermediate Antigravity CLI, Codex, git, OpenFOAM, LLMs.

Advanced C/C++, UNIX/Linux [openSUSE, (K)Ubuntu], TEX/LATEX, MATLAB, Python.

Languages

Vietnamese Mother tongue

English Good

Teaching

Mathematics Discrete Mathematics, Linear Algebra, Introduction to Mathematical Analysis, Combinatorics & Graph Theory (Summer 2025, Summer 2026).

Computer Science Fundamentals of Programming (C/C++), Information Technology Fundamentals II (Python), Machine Learning with Python (Summer 2025, Summer 2026).

Supervision Bachelor Thesis Supervision in Mathematics/Computer Science (Spring 2026).

Students

In alphabetical order:

[1] Nguyễn Ngọc Thạch (2026) (Software Engineering). *Chronix: Personalized Scheduling with AI and Chronobiology for Optimal Productivity – Chronix: Hệ Thống Lập Lịch Cá Nhân Hoá Tối Ưu Năng Suất Dựa Trên AI và Nhịp Điệu Sinh Học*. Current: Software Engineer at Zalo, VNG Corporation, Vietnam. [pdf]

Referees

In chronological order:

[1] **Dr. Đào Mạnh Khang** — *Bachelor Teacher*

[2] **Dr. Nguyễn Tấn Trung** — *Bachelor Teacher, email: trung.nguyentan@umt.edu.vn.*

[3] **Dr. Đào Nguyên Anh** — *Bachelor Supervisor*

[4] **Prof. Luis Miguel Rodrigues** — *Master 2 Program Chair, email: luis-miguel.rodrigues@univ-rennes.fr.*

[5] **Prof. Dr. Trần Đan Thư** — *Head of IT Department of UMT, email: : thu.tran@umt.edu.vn.*

Hobbies

- Psychology Ψ
- Philosophy Φ
- Literary
- Writing, blogging

Updated to Sep 30, 2026